EcoNaija Manufacturing Ltd: Carbon Footprint & Decarbonisation Strategy

Executive summary · 04 November 2026 · Prepared by Daniel Oyewale, Petgrave.io

Illustrative case study. Activity data and company profile are synthetic; emission factors (DESNZ 2026, verified) and 2026 Nigerian energy prices are real. Not a verified GHG inventory.

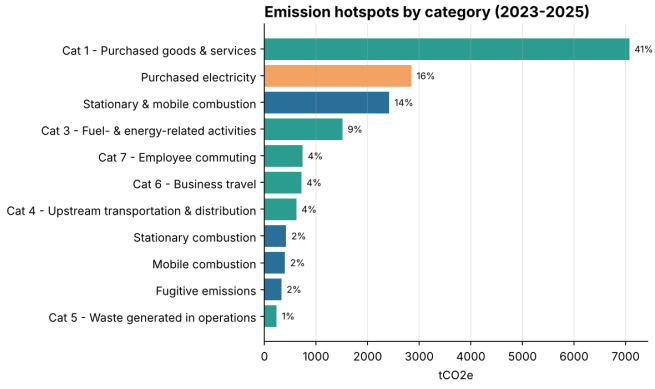
Headline

2023-25 footprint	2025 emissions	Change vs 2023	Modelled abatement	Net saving	2030 SBTi gap*
17,313 t	**5,384 t**	**-12.3%**	**967 t/yr**	**$124k/yr**	**-174 t**

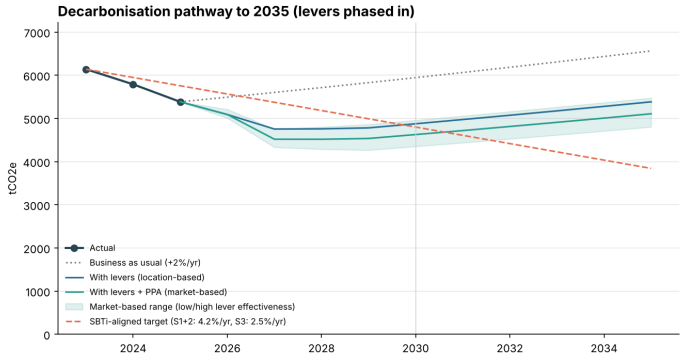
tCO2e. *Market-based, central case, vs SBTi-aligned target (Scope 1+2 –4.2%/yr, Scope 3 –2.5%/yr from 2023); negative = on track.

What the footprint shows

- **Scope 3 dominates (63%).** Cat 1 - Purchased goods & services is the largest single source (41%), led by steel (18%) and plastic resin (12%). Supplier data on these materials is the biggest driver of accuracy and of future reductions.
- **Energy is the controllable core.** Grid electricity (Scope 2, 16%) and diesel generation (Scope 1) concentrate at Port Harcourt Plant, which is 56% of the total.
- **Emissions fell 12.3% from 2023 to 2025** (real terms), ahead of a straight-line path, but without new measures 2%/yr business growth would push them back up.



Synthetic data: EcoNaija Manufacturing Ltd (fictional). Factors: DESNZ 2026, derived Nigeria grid (NERC 2025), US EPA v1.3.



Lever costs are modelled assumptions (see data/reference/*.csv). Synthetic data: EcoNaija Manufacturing Ltd (fictional). Factors: DESNZ 2026, derived Nigeria grid (NERC 2025), US EPA v1.3.

Recommended investment plan

Priority	Lever	tCO2e/yr	Capex	USD/tCO2e	Payback
Fund now	Travel policy (virtual-first + economy-only)	67	$15k	-812	0.3 yr
Fund now	Solar-battery hybrid at Port Harcourt Plant	266	$216k	-293	2.0 yr
Fund now	Energy efficiency (LED + inverter ACs + motor upgrades)	140	$70k	-190	1.8 yr
Fund now	Rooftop solar PV (grid-tied) at Lagos HQ & Abuja Office	112	$111k	-130	3.8 yr
Next	Hybrid working policy (2 remote days per week)	83	$0k	0	–
Next	Renewable PPA / I-RECs for Port Harcourt Plant grid supply	227	$0k	48	–
Next	Refrigerant leak detection & maintenance programme	62	$8k	61	–
Strategic	Waste segregation & recycling programme	37	$4k	116	–
Strategic	Supplier engagement programme (top 20 suppliers)	201	$0k	149	–

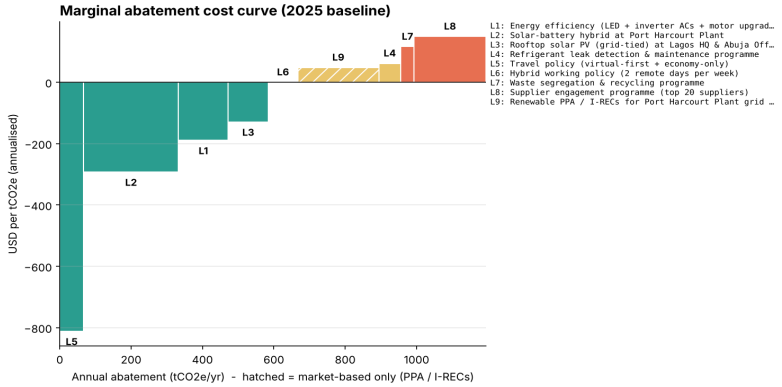
Decision points for the board

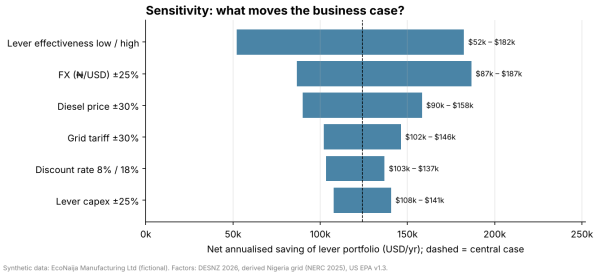
1. Approve the 4 cost-negative levers now. They deliver 49% of modelled abatement and pay for themselves. The flagship is the **Port Harcourt solar-battery hybrid**: 266 tCO2e/yr, ~$216k capex, ~2-year payback, plus protection from diesel price shocks.

2. Contract renewable electricity for the plant (PPA / I-RECs). Without it, 2030 is missed on a location basis (+77 t). With it, the central case is on track (-174 t). Certificates must pass the GHG Protocol Scope 2 Quality Criteria (same-year vintage, Nigerian market, retired in a registry).

3. Plan a second wave by ~2029. If levers underperform, 2030 slips to +159 t, and growth erodes gains toward 2035. Candidates: deeper supplier engagement, fleet conversion (CNG/EV), expanded on-site solar.

4. Invest in data before targets are published. Collect supplier-specific emissions data for the top spend categories (steel, resin) and meter diesel use by generator. These are the largest sources of uncertainty.





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Robustness: the portfolio stays cash-positive in every sensitivity case tested (minimum ~$52k/yr). Lever effectiveness and the NGN/USD exchange rate move the business case most.

Method, sources & limitations

- GHG Protocol Corporate & Scope 3 Standards; operational control; 2023 base year; AR5 GWP100.
- Scope 1 & travel/waste factors: **DESNZ 2026**, verified cell-by-cell against the official workbook. 100% mineral fuel factors (no biofuel blend in Nigerian AGO/PMS).
- Nigeria grid: **Scope 2 combustion-only factor derived** from the NERC 2025 generation mix (67% gas), DESNZ gas combustion factor and a 33% fleet efficiency (range 28-38%). Upstream fuel supply and T&D losses are reported separately in Scope 3 Category 3.
- Scope 3 spend: 10 categories mapped to **US EPA Supply Chain Factors v1.3** (NAICS-6, with margins), with nominal spend deflated to 2022 USD using US CPI-U. Freight is reported as Category 4.
- Market-based Scope 2 only counts contractual instruments that pass the GHG Protocol Scope 2 Quality Criteria; Nigeria has no residual mix, so the grid average is the fallback.
- Scenario costs use Oct 2026 Nigerian prices (diesel NGN 1,900/L, Band A NGN 209.5/kWh, NGN 1,350/USD, 12% discount rate). Lever capex/opex are documented assumptions, phased in over 1-3 years with low/high effectiveness ranges.
- Limitations: synthetic activity data; spend-based Scope 3 uses US-economy factors; UK-based proxies for landfill and car factors likely understate Nigerian conditions; no grid decarbonisation assumed.

Full model, code and data: github.com/Engr-Daniel/carbon-footprint-esg-dashboard